

Marlin EV front brake hose internal swelling causing dragging caliper

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Area: bulletins

1 Condition

Affected Marlin EV vehicles may exhibit a dragging front brake caliper, assembly BCAL-7760, accompanied by uneven pad wear, localized wheel heat, and reduced range. The condition typically appears after the vehicle has been parked following a braking event, as residual hydraulic pressure fails to release. Technicians may observe that the BCAL-7760 caliper piston does not fully retract when the brake pedal is released. For caliper inspection criteria and torque values, refer to the Marlin EV Brakes Service Manual :: Caliper Sub-System .

2 Cause

The root cause is internal swelling of the front brake hose, part BH-7745, in which the inner liner delaminates and acts as a one-way check valve. This allows hydraulic pressure to reach the caliper but restricts return flow, holding the caliper applied. The defective BH-7745 hose was produced within a limited supplier date range and should be replaced rather than reused. See Marlin EV Brakes Service Manual :: Hydraulic Lines and Fluid for hose routing and inspection guidance.

3 Repair Procedure: Brake Hose Replacement

Replace the affected front brake hose BH-7745 following procedure PROC-BRAKE-HOSE, ensuring the new hose is routed without twist and seated against the chassis bracket. Support the caliper during disconnection and cap the open line to minimize fluid loss and air ingress. The complete step-by-step replacement is documented in the Marlin EV Brakes Service Manual :: Brake Hose Replacement Procedure . After fitting BH-7745, proceed to caliper service and a full system bleed.

4 Caliper Service and Brake Fluid Bleed

Inspect caliper BCAL-7760 for piston seizure or scoring; if the piston does not retract freely after the hose is replaced, replace the caliper per PROC-CALIPER as described in the Marlin EV Brakes Service Manual :: Brake Caliper Replacement Procedure . Once BCAL-7760 is serviced, flush and bleed the system using brake fluid BF-7790 to remove any contaminated or aerated fluid. Bleed in the manufacturer-specified sequence and verify a firm pedal before release. The complete fluid specification and bleed sequence appear in the Marlin EV Brakes Service Manual :: Brake Fluid Flush and Bleed Procedure .